

THE CONGRUENCE SUBGROUP PROPERTY

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1. $\mathrm{SL}_n\mathbb{Z}$

1.1. Congruence subgroups.

Definition 1.1.1 (Principal congruence subgroup). Let $\mathcal{R}_m : \mathrm{SL}_n(\mathbb{Z}) \rightarrow \mathrm{SL}_n(\mathbb{Z}/m)$ be the homomorphism induced by reduction modulo m . We call $\ker \mathcal{R}_m$ the principal congruence subgroup of level m in $\Gamma = \mathrm{SL}_n(\mathbb{Z})$, and we denote it by $\Gamma(m)$.

When $n = 2$, we have the following description for $\Gamma(m)$.

$$\Gamma(m) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma : a, d \equiv 1 \pmod{m}, b, c \equiv 0 \pmod{m} \right\}.$$

Remark 1.1.2. The Chinese Remainder Theorem implies that $\mathcal{R}_m$ is surjective, so $\Gamma/\Gamma(m) \cong \mathrm{SL}_2(\mathbb{Z}/m)$.

Definition 1.1.3 (Congruence subgroup). A subgroup $H \leq \Gamma$ is called a congruence subgroup if there exists $m \geq 1$ such that $\Gamma(m) \leq H$.

The level l of H is defined as the smallest such n .

1.2. The congruence subgroup problem for $\mathrm{SL}_n\mathbb{Z}$.

Definition 1.2.1 (Congruence subgroup property). We say that Γ enjoys the congruence subgroup property if any finite index subgroup of Γ is congruence.

Theorem 1.2.2. $\Gamma = \mathrm{SL}_2\mathbb{Z}$ does not enjoy the congruence subgroup property.

One way to obtain non-congruence subgroups is via the kernel of $\Gamma(2) \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/m$.

Remark 1.2.3. Every normal subgroup of $\mathrm{PSL}_2(\mathbb{Z})$ is free, apart from three exceptions. Similarly, every normal subgroup of $\mathrm{SL}_2(\mathbb{Z})$ which does not contain $-I$ must be free.

Theorem 1.2.4. For $n \geq 3$, $\mathrm{SL}_n\mathbb{Z}$ enjoys the congruence subgroup property.

2. MAPPING CLASS GROUPS

2.1. Definitions and motivation.

Recall the isomorphism $\mathrm{MCG}(T^2) \cong \mathrm{SL}_2\mathbb{Z}$ coming from the action of $\mathrm{MCG}(S_g)$ on $H_1(S_g; \mathbb{Z})$, which has kernel $\mathcal{I}(S_g)$, the Torelli group. This is,

$$\mathcal{I}(S_g) = \ker(\mathrm{MCG}(S_g) \rightarrow \mathrm{Aut} H_1(S_g; \mathbb{Z}))$$

$$\mathcal{I}(S_g) = \ker(\mathrm{MCG}(S_g) \rightarrow \mathrm{Sp}_{2g}\mathbb{Z}).$$

When $g = 1$, this gives the isomorphism $\mathrm{MCG}(T^2) \cong \mathrm{SL}_2\mathbb{Z}$.

Definition 2.1.1 (Principal congruence subgroup of level m). We define the principal congruence subgroup of level m of $\mathrm{MCG}(S_g)$ as the kernel of the composition $\mathrm{MCG}(S_g) \rightarrow \mathrm{Sp}_{2g}(\mathbb{Z}) \rightarrow \mathrm{Sp}_{2g}(\mathbb{Z}/m)$. We write such kernel as $\mathrm{MCG}(S_g)[m]$.

Remark 2.1.2.

$$\mathcal{I}(S_g) \leq \text{MCG}(S_g)[m].$$

Question 1. Does every finite index subgroup of $\text{MCG}(S)$ contain a principal congruence subgroup of level m for some m ?

The answer is no. It suffices to find subgroups of finite index that do not contain $\mathcal{I}(S_g)$, e.g the Morita subgroups.

2.2. Dehn-Nielsen-Baer.

Let $\text{Out}^*(\pi_1(S)) \leq \text{Out}(\pi_1(S))$ consisting of elements that preserve the set of conjugacy classes of the simple closed curves surrounding individual punctures.

Theorem 2.2.1. Let $S = S_{g,n}$ be hyperbolic. Then

$$\text{MCG}^\pm(S) \cong \text{Out}^*(\pi_1(S)).$$

Definition 2.2.2 (Characteristic subgroup). A subgroup $N \leq G$ is characteristic if it is invariant under every automorphism of G .

Definition 2.2.3 (Geometric automorphism). An automorphism of $\pi_1(S)$ is said to be geometric if it is induced by a homeomorphism of S .

Example 2.2.4. $S = S_{1,2}$. Note that $\pi_1(S) \cong F_3$. Take a generator a for $\pi_1(S)$ along a separating simple closed curve, and another generator b along a separating curve. Then the automorphism induced by $a \mapsto b$ is not geometric.

Theorem 2.2.5 (Dehn-Nielsen-Baer).

$$\text{MCG}^\pm(S_g) \cong \text{Out}(\pi_1(S_g))$$

Definition 2.2.6. Let $S = S_{g,n}$, and let $H \leq \pi_1(S)$ be a finite index geometrically characteristic subgroup, and consider the map

$$\phi_H : \text{Out}^*(\pi_1(S_{g,n})) \rightarrow \text{Out}(\pi_1(S_{g,n})/H).$$

Then we call $\ker \phi_H$ a principal congruence subgroup.

Definition 2.2.7 (Congruence subgroup of mapping class groups). Let $H \leq \pi_1(S_g)$ be a characteristic subgroup of finite index. We have a homomorphism

$$\phi_H : \text{MCG}(S_g) \rightarrow \text{Out}(\pi_1(S_g)/H).$$

We call $\ker \phi_H$ a principal congruence subgroup.

Question 2. Does every finite index subgroup of $\text{MCG}(S)$ contain a principal congruence subgroup?

Proposition 2.2.8. Let S be an orientable hyperbolic surface of finite type. Then $\text{PMCG}(S)$ is congruence in $\text{MCG}(S)$.

Proof. We show that $\text{MCG}(S_{g,n})[m] \leq \text{PMCG}(S_{g,n})$ for all $m \geq 2$. ■

Lemma 2.2.9. If $\text{PMCG}(S_{g,n-1})$ has the CSP, then $\text{PMCG}(S_{g,n})$ has the CSP.

Lemma 2.2.10. From the Birman Exact Sequence

$$1 \rightarrow \pi_1(S_{g,n-1}) \xrightarrow{\mathcal{P}} \text{PMCG}(S_{g,n}) \xrightarrow{\mathcal{F}} \text{PMCG}(S_{g,n-1}) \rightarrow 1,$$

$\mathcal{P}\pi_1(S_{0,n-1})$ has the CSP.

Proposition 2.2.11. $\text{PMCG}(S_{0,n})$ has the CSP.

Question 3. Is every hyperbolic group residually finite?

Theorem 2.2.12 (Wilton, 2024). If every hyperbolic group is residually finite, then $\text{MCG}(S_{g,n})$ enjoys the congruence subgroup property.